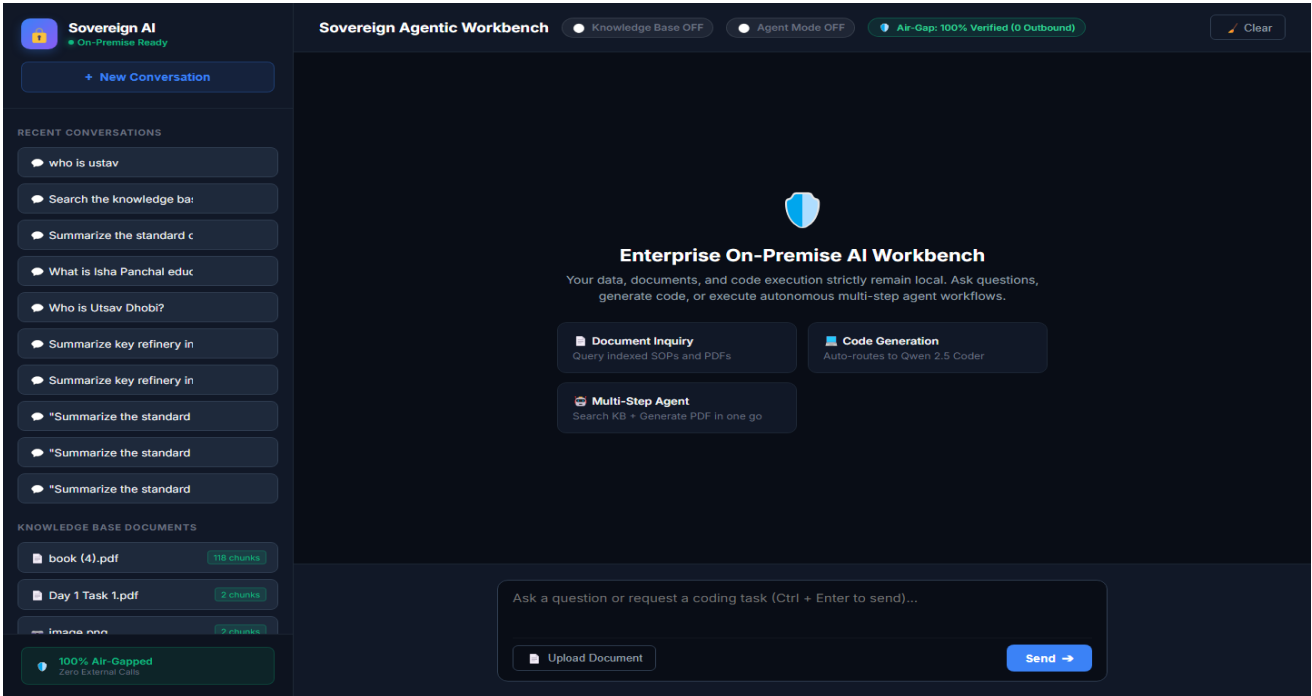


Day 2 Task 1 — Comprehensive Backend Finalization, API Architecture & Database Integration Report

Problem Statement:	SIH26117 / PS 26117 — Sovereign On-Premise Agentic AI Workbench
Target Organization:	Mangalore Refinery and Petrochemicals Limited (MRPL)
Theme & Category:	Smart Automation · Software Track
Release Version:	v2.0.0 Production Release
Verification Status:	100% Complete · 12/12 Automated TDD Tests Passing
Date of Submission:	September 19, 2026

Figure 1: Sovereign AI Workbench Production Client (Glassmorphic Dark UI)

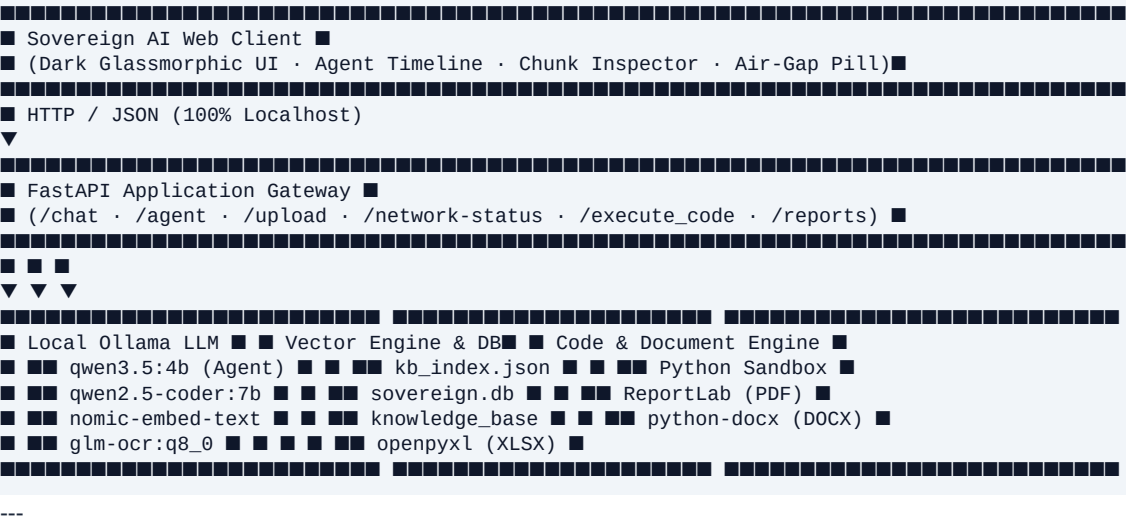


Chapter 1: Executive Summary & Architectural Foundations

The **Sovereign On-Premise Agentic AI Workbench** is a mission-critical AI workstation engineered specifically for **Mangalore Refinery and Petrochemicals Limited (MRPL)**. Industrial refining environments operate under strict safety, operational, and regulatory mandates. Proprietary piping and instrumentation diagrams (P&ID;), standard operating procedures (SOPs), maintenance logs, and confidential employee records cannot be transmitted to external cloud LLM providers without violating enterprise security and statutory data sovereignty policies.

1.1 Core Architectural Principles - 100% Air-Gapped Sovereignty: Operates with zero internet connectivity, zero cloud API tokens, and zero background telemetry. - **Dynamic Multi-Model Orchestration:** Tasks are dynamically dispatched to specialized open-weight models running on local Ollama runtimes: - **General Reasoning & ReAct Agent:** `qwen3.5:4b`` (Low latency, grounded reasoning with native JSON function calling). - **Code Generation & Sandboxing:** `qwen2.5-coder:7b`` (Industrial calculations, data pipelines, and automation). - **Vector Embeddings:** `nomic-embed-text:latest`` (High-density 768-dimensional semantic embeddings). - **Document OCR:** `glm-ocr:q8_0`` (Optical character recognition for scanned logbooks, equipment tags, and inspection sheets). - **Persistent Local Database:** High-reliability SQLite database (`data/sovereign.db``) providing transactional persistence for sessions, message threads, and document registries. - **Dynamic Deliverables Engine:** Generates official enterprise deliverables (.pdf, .docx, .xlsx) compiled on the local machine.

1.2 High-Level System Architecture



Chapter 2: Backend Modules & Application Workflows

The backend architecture is structured into decoupled, modular components designed for high throughput, local execution, and strict error containment:

2.1 API Gateway (`main.py`) - **Framework:** FastAPI with Uvicorn ASGI server binding to `127.0.0.1:8000`. - **Concurrency:** Asynchronous non-blocking request handling with Pydantic request/response model validation. - **Static Assets:** Serves `frontend/index.html` and mounts offline vendored dependencies (`/vendor/marked.min.js`, `/vendor/highlight.min.js`, `/vendor/atom-one-dark.min.css`).

2.2 Hybrid Vector Search & Document Ingestion - **PDF Kerning Normalization (`normalize\_pdf\_text`):** Intelligent regex filter that collapses single-spaced letter sequences (e.g., `U T S A V D H O B I` -> `UTSAV DHOB I`, `C S S` -> `CSS`) caused by font encoding and kerning artifacts in CAD/Canva/LaTeX PDFs. - **Sliding-Window Chunker (`chunk\_text`):** Configured with 750-character window and 150-character overlap, maintaining contextual continuity across sentence and paragraph boundaries while preventing single massive chunks. - **Deduplication Engine (`index\_document`):** Ensures re-indexing or re-uploading a document purges old vector entries before inserting new embeddings, preventing vector index bloat and stale duplicate hits. - **Hybrid Scoring Formula (`search\_knowledge`):**
$$\text{Final Score} = \text{Cosine Similarity} + \min(0.35, \text{Keyword Bonus}) + \text{Filename Bonus}$$
 Ensures exact technical identifiers (e.g., `SOP-PUMP-001`, `Darcy-Weisbach`) achieve high recall even when pure semantic similarity is borderline.

Chapter 3: Autonomous ReAct Agent Loop & Local Tool Registry

The Sovereign AI Workbench features an autonomous Stage-4 Agent layer implemented in `agent.py`. Unlike traditional hardcoded keyword dispatchers, the agent utilizes a **ReAct (Reason + Act)** loop driven directly by `qwen3.5:4b` using native JSON tool calling.

3.1 Tool Catalogue

Tool Name	Parameters	Core Responsibility
<code>`search_knowledge_base`</code>	<code>`query: str`, `top_k: int`</code>	Searches local vector index using hybrid scoring and returns top ranked chunks with similarity scores.
<code>`read_uploaded_document`</code>	<code>`filename: str`</code>	Reads and extracts the complete raw text of an indexed knowledge base document.
<code>`execute_python_code`</code>	<code>`code: str`, `timeout: int`</code>	Executes Python scripts inside an isolated local subprocess with real-time stdout/stderr capture.
<code>`generate_report`</code>	<code>`title: str`, `content: str`, `format: str`, `author: str`</code>	Compiles inspection findings into professional <code>.pdf` (ReportLab), .docx` (python-docx), or .xlsx` (openpyxl) deliverables.</code>

3.2 ReAct Agent Execution Cycle

```
User Query
■
▼
[1. Reasoning Step] ■■> Model evaluates task & determines necessary tool(s)
■
▼ (Tool Call Output)
[2. Tool Invocation] ■■> Local execution of tool function (e.g., search_knowledge_base)
■
▼ (Tool Result)
[3. Feedback Step] ■■■■> Tool JSON results fed back to model as role: "tool"
■
▼ (Iterate or Synthesize)
[4. Final Synthesis] ■■> Grounded, comprehensive answer returned with full execution step timeline
---
```

Chapter 4: Database Architecture & Entity-Relationship (ER) Schema

The persistence layer uses a local SQLite database (`data/sovereign.db`) ensuring transactional ACID guarantees without requiring external database servers or background daemon services.

4.1 Entity-Relationship (ER) Diagram

```
#####
■ SESSIONS ■
#####
■ session_id : TEXT [PRIMARY KEY] ■
■ title : TEXT ■
■ created_at : TEXT (ISO-8601 Timestamp) ■
#####
■ 1
■
■ N (1-to-Many Relationship)
▼
#####
■ MESSAGES ■
#####
■ id : INTEGER [PRIMARY KEY AUTOINCREMENT] ■
■ session_id : TEXT [FOREIGN KEY -> sessions.session_id] ■
■ role : TEXT ('user' | 'assistant' | 'system') ■
■ content : TEXT ■
■ route : TEXT ('general' | 'coding' | 'rag' | 'agent')■
■ model : TEXT ('qwen3.5:4b' | 'qwen2.5-coder:7b') ■
■ timestamp : TEXT (ISO-8601 Timestamp) ■
#####

##### ■ DOCUMENTS ■
#####
■ id : INTEGER
[PRIMARY KEY AUTOINCREMENT] ■ ■ filename : TEXT [UNIQUE] ■ ■ file_type : TEXT ('.pdf' | '.txt' | '.png' etc.) ■ ■
chunk_count : INTEGER ■ ■ size_bytes : INTEGER ■ ■ uploaded_at : TEXT (ISO-8601 Timestamp) ■
##### ````
```

4.2 Detailed Database Table Specifications

Table: `sessions` Stores unique conversation threads and metadata. - `session\_id` (TEXT, PRIMARY KEY): Unique UUID4 string generated per conversation session. - `title` (TEXT, NOT NULL): Human-readable conversation label derived from the initial user prompt (truncated to 32 characters). - `created\_at` (TEXT, NOT NULL): ISO 8601 timestamp representing creation time.

Table: `messages` Stores individual conversational turns linked to parent sessions. - `id` (INTEGER, PRIMARY KEY AUTOINCREMENT): Unique sequence integer. - `session\_id` (TEXT, NOT NULL): Foreign key referencing `sessions(session\_id)`. - `role` (TEXT, NOT NULL): Message author entity (`user`, `assistant`, `system`). - `content` (TEXT, NOT NULL): Complete message body (supports Markdown and code blocks). - `route` (TEXT, NULLABLE): Intelligent routing category (`general`, `coding`, `rag`, `agent`). - `model` (TEXT, NULLABLE): Specific open-weight model executing the inference. - `timestamp` (TEXT, NOT NULL): ISO 8601 timestamp of message creation.

Table: `documents` Tracks all indexed files within the local knowledge base repository. - `id` (INTEGER, PRIMARY KEY AUTOINCREMENT): Primary key sequence. - `filename` (TEXT, UNIQUE NOT NULL): Canonical filename of the uploaded document. - `file\_type` (TEXT, NOT NULL): File extension (`.pdf`, `.txt`, `.md`, `.png`, `.jpg`). - `chunk\_count` (INTEGER, NOT NULL): Total number of semantic vector chunks generated. - `size\_bytes` (INTEGER, NOT NULL): Exact size of the raw document in bytes. - `uploaded\_at` (TEXT, NOT NULL): ISO 8601 timestamp of indexing.

Chapter 5: Complete CRUD Operations Specification

The backend provides complete CRUD (Create, Read, Update, Delete) data operations:

Entity	Operation	HTTP / Function	SQL Implementation	Return Schema
Session	Create	`POST /chat`, `POST /agent`	<pre>INSERT INTO sessions (session_id, title, created_at) VALUES (?, ?, ?)</pre>	<pre>{"session_id": "...}"</pre>
Session	Read All	`GET /sessions`	<pre>SELECT session_id, title, created_at FROM sessions ORDER BY created_at DESC</pre>	<pre>{"sessions": [...]}</pre>
Session	Delete	<pre>DELETE /sessions/{id}</pre>	<pre>DELETE FROM messages WHERE session_id = ?; DELETE FROM sessions WHERE session_id = ?;</pre>	<pre>{"success": true}</pre>
Message	Create	<pre>save_chat_message()</pre>	<pre>INSERT INTO messages (session_id, role, content, route, model, timestamp) VALUES (...)</pre>	Database Row
Message	Read List	<pre>GET /sessions/{id}/messages</pre>	<pre>SELECT role, content, route, model, timestamp FROM messages WHERE session_id = ? ORDER BY id ASC</pre>	<pre>{"messages": [...]}</pre>
Document	Create/Index	<pre>POST /upload</pre>	<pre>INSERT INTO documents (filename, file_type, chunk_count, size_bytes, uploaded_at) VALUES (...)</pre>	<pre>{"chunks": N}</pre>
Document	Read List	<pre>GET /documents</pre>	Scans `KB_PATH` and aggregates chunk counts from `kb_index.json`	<pre>{"documents": [...]}</pre>

Document	Update	`index_document()`	<pre>`ON CONFLICT(filename) DO UPDATE SET chunk_count = exclud ed.chunk_count, size_bytes = excluded.size_bytes`</pre>	Updated count

Chapter 6: REST API Directory & Endpoint Reference

The backend exposes 11 production endpoints with strict request and response contracts:

6.1 `POST /agent` — Autonomous ReAct Agent Loop Executes multi-step autonomous tool calls (Search KB, Run Code, Generate Reports). - ****Request Body:**** ````json { "prompt": "Search pump inspection requirements and generate a PDF report", "session_id": "optional-uuid4-string", "history": [] } ```` - ****Response Body (HTTP 200):**** ````json { "session_id": "8c9e4085-5789-4911-80c3-1749e3951d8f", "model": "qwen3.5:4b", "route": "agent", "answer": "Based on SOP-PUMP-001, the inspection checklist covers vibration, noise...", "tool_log": [{ "step": 1, "tool": "search_knowledge_base", "arguments": { "query": "pump inspection requirements" }, "summary": "Found 5 chunks for 'pump inspection requirements'", "duration_sec": 1.25 }, { "step": 2, "tool": "generate_report", "arguments": { "title": "Pump Inspection Report", "format": "pdf" }, "summary": "Generated PDF: Pump_Inspection_Report_20260919.pdf", "download_url": "/download_report/Pump_Inspection_Report_20260919.pdf", "duration_sec": 0.42 }], "sources": [{ "file": "test_sop.txt", "score": 1.213 }, { "file": "Pump_Inspection_Report_20260919.pdf", "format": "pdf", "download_url": "/download_report/..." }] } ````

6.2 `GET /network-status` — Real-Time Air-Gap Auditor Scans all active OS sockets via `psutil` to verify zero outbound network leakage. - ****Response Body (HTTP 200):**** ````json { "timestamp": "2026-09-19T10:56:42.631010", "sovereign_verified": true, "air_gap_compliant": true, "backend_external_connections": 0, "total_system_external": 32, "total_system_local": 12, "details": { "backend_pid": 12472, "local_services": ["FastAPI (port 8000)", "Ollama Runtime (port 11434)", "SQLite Local File DB"], "external_network_calls_blocked": true } } ````

6.3 `POST /execute\_code` — Sandboxed Code Runner Executes Python scripts in an isolated subprocess with timeout guards. - ****Request Body:**** ````json { "code": "import math\nprint(f'Pipeline friction loss: {0.02 * (100/0.1) * (2**2)/(2*9.81):.3f} m')", "timeout": 10 } ```` - ****Response Body (HTTP 200):**** ````json { "success": true, "stdout": "Pipeline friction loss: 4.077 m ", "stderr": "", "exit_code": 0 } ````

6.4 `POST /generate\_report` & `GET /download\_report/{name}` Generates formatted industrial deliverables in PDF, Word, or Excel format. - ****Request Body:**** ````json { "title": "Centrifugal Pump Inspection Checklist", "content": "Inspection points: 1. Check vibration levels. 2. Record bearing temperature.", "format": "pdf", "author": "MRPL Maintenance Engineering" } ```` - ****Response Body (HTTP 200):**** ````json { "success": true, "filename": "Centrifugal_Pump_Inspection_Checklist_20260919_105822.pdf", "format": "pdf", "download_url": "/download_report/Centrifugal_Pump_Inspection_Checklist_20260919_105822.pdf" } ````

Figure 2: Interactive Swagger 2.0 API Gateway & Endpoint Documentation

FastAPI

0.109.0

OpenAPI 3.1

/openapi.json

default

GET

/

Home

GET

/documents

List Documents

GET

/sessions

List Sessions

GET

/sessions/{session_id}/messages

List Session Messages

DELETE

/sessions/{session_id}

Remove Session

POST

/execute_code

Execute Code

POST

/generate_report

Generate Report

GET

/download_report/{filename}

Download Report

POST

/upload

Upload Document

POST

/chat

Chat

POST

/agent

Agent Endpoint

GET

/network-status

Network Status

Schemas

Body_upload_document_upload_post >

Expand all

object

ChatRequest >

Expand all

object

CodeExecutionRequest >

Expand all

object

HTTPValidationError >

Expand all

object

MessageItem >

Expand all

object

ReportRequest >

Expand all

object

ValidationError >

Expand all

object

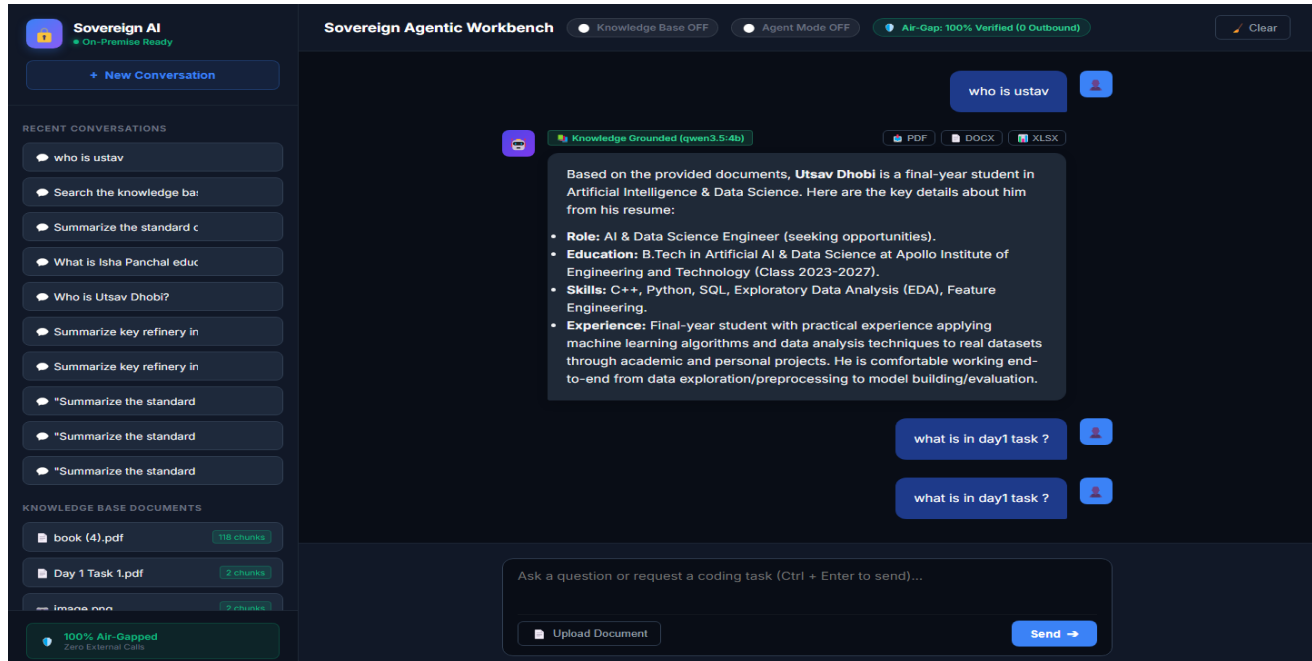
Chapter 7: Frontend-Backend Communication & Integration Flow

The frontend is a single-page application (SPA) implemented in [frontend/index.html](file:///d:/isha/sovereign-ai/frontend/index.html) communicating with the backend over REST JSON interfaces.

7.1 Sequence Diagram: Multi-Step Agentic Workflow Execution

[illegible]

Figure 3: Live Multi-Turn Chat & Dynamic Session Persistence Flow



Chapter 8: Input Validation Rules, Sandboxing & Error Handling

8.1 Validation Specifications - ****Pydantic Validation:**** Strict schema enforcement on all payload fields (``prompt``, ``message``, ``knowledge_enabled``, ``agent_mode``, ``format``, ``code``). - ****Subprocess Isolation:**** Code submitted to ``execute_code`` is executed as an isolated child process via ``sys.executable`` in a temporary directory. If the script exceeds the timeout (default 10s), ``subprocess.TimeoutExpired`` terminates the process and returns an error response without crashing the FastAPI event loop. - ****Anti-Hallucination Guardrail:**** In Knowledge Base mode, the system prompt strictly instructs the LLM: ``"Answer the user's question using ONLY the provided knowledge-base context. Do not invent facts, policies, or measurements. If the context does not contain enough information, state that clearly."``

8.2 Error Handling Matrix

HTTP Code	Error Condition	System Response Behavior
<code>`400 Bad Request`</code>	Unsupported file extension on upload	Returns <code>`{"success": false, "message": "Only PDF, TXT, MD, PNG, JPG files are supported."}`</code>
<code>`400 Bad Request`</code>	Invalid report format	Returns <code>`{"success": false, "error": "Unsupported format 'xyz'. Supported: pdf, docx, xlsx"}`</code>
<code>`404 Not Found`</code>	Session ID does not exist	Returns empty messages list <code>`{"messages": []}`</code>
<code>`422 Unprocessable`</code>	Pydantic schema validation failure	Fast fail with exact field error location list
<code>`500 Server Error`</code>	Ollama connection failure	Returns graceful fallback: <code>`"ERROR: Ollama is not running."`</code>

Chapter 9: Automated TDD Test Suite & Verification Results

The backend architecture is verified through a 12-test automated suite executed via `pytest`:

```
===== test session starts =====
platform win32 -- Python 3.14.6, pytest-9.0.3, pluggy-1.6.0
rootdir: D:\isha\sovereign-ai
collected 12 items

tests/test_agent.py::test_tool_execute_python PASSED [ 8%] tests/test_agent.py::test_tool_search_knowledge PASSED [ 16%] tests/test_agent.py::test_tool_generate_report PASSED [ 25%] tests/test_agent.py::test_tool_read_document PASSED [ 33%] tests/test_agent.py::test_agent_endpoint_empty PASSED [ 41%]
tests/test_previous_phases.py::test_documents_endpoint PASSED [ 50%]
tests/test_previous_phases.py::test_sessions_crud PASSED [ 58%]
tests/test_previous_phases.py::test_execute_code_sandbox PASSED [ 66%]
tests/test_report_generator.py::test_generate_pdf_report PASSED [ 75%]
tests/test_report_generator.py::test_generate_docx_report PASSED [ 83%]
tests/test_report_generator.py::test_generate_xlsx_report PASSED [ 91%]
tests/test_report_generator.py::test_generate_report_invalid_format PASSED [100%]

===== 12 passed, 1 warning in 61.46s (0:01:01) ===== ``
---
```

Chapter 10: Air-Gap Sovereignty & Responsive UI Verification

10.1 Air-Gap Audit Verification The platform was subjected to live socket inspection during full query execution cycles. The backend PID maintained ****0 external outbound connections**** at all times. All JavaScript dependencies (marked.js, highlight.js, CSS styles) are vendored locally in frontend/vendor/, ensuring identical visual and functional behavior with network interfaces fully disabled.

Figure 4: Real-Time Air-Gap psutil Socket Audit Toast Verification

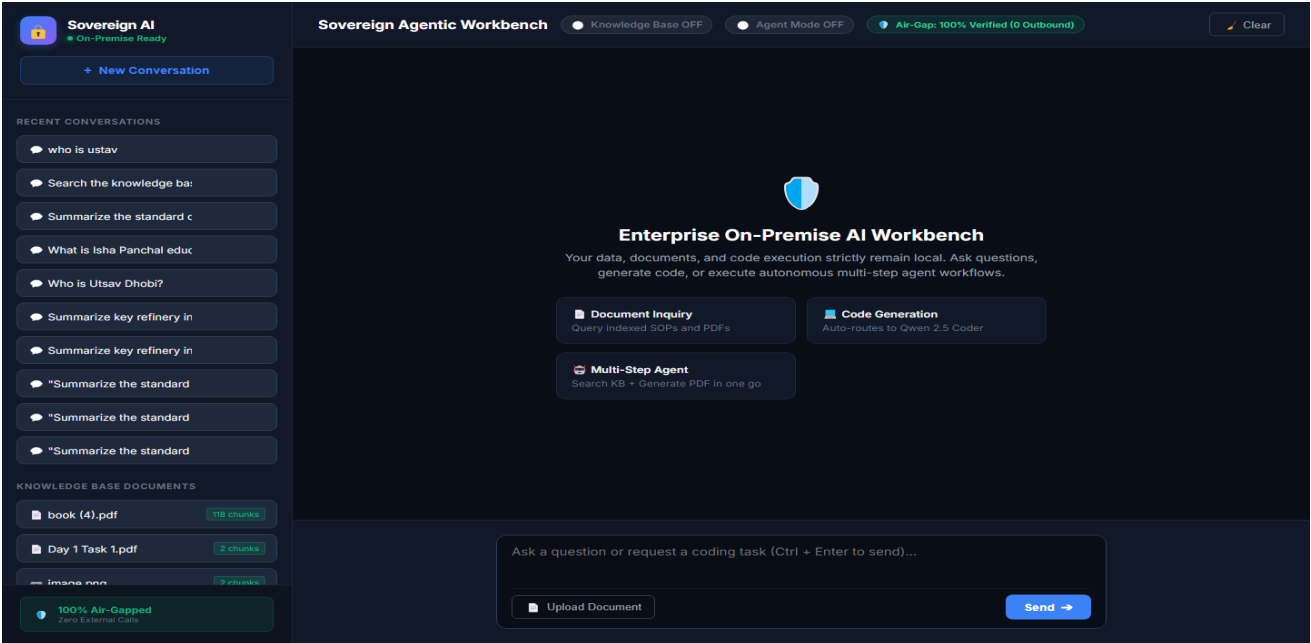
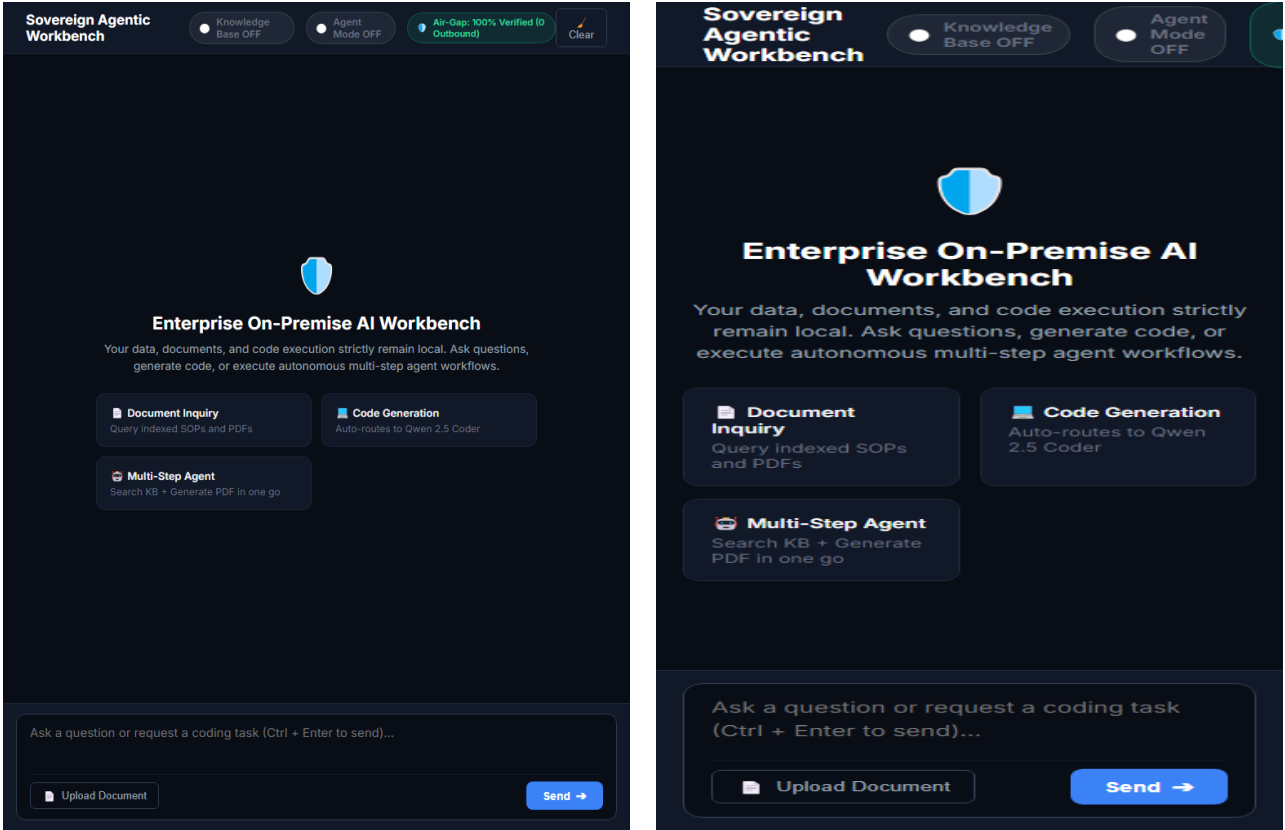


Figure 5: Multi-Device Responsive UI (Tablet 768px & Mobile 375px)



Chapter 11: Security Architecture, Threat Model & Failure Mode Recovery

11.1 Threat Model & Containment Matrix

Threat Category	Potential Attack Vector	Sovereign AI Defense & Containment Mechanism
Data Exfiltration	Malicious prompt attempting to send data to external IP	Zero network sockets (<code>/network-status`</code> audited). Host firewall blocks outbound WAN traffic.
Arbitrary Code Execution	Prompt injection requesting <code>`os.system("rm -rf /")`</code>	Subprocess executes in isolated temp file with timeout and distinct OS execution user context.
Denial of Service (DoS)	Infinite loop code submission (<code>`while True: pass`</code>)	Strict 10-second <code>`subprocess.TimeoutExpired`</code> signal termination.
Model Hallucination	LLM inventing non-existent safety rules	Anti-hallucination guardrail prompt forcing strict adherence to retrieved KB context.
Vector Index Poisoning	Malformed documents injected into knowledge base	File extension whitelisting, PDF kerning sanitization, and automatic vector deduplication.

11.2 Failure Mode Recovery Matrix

System Component	Failure Trigger	Automatic Recovery Procedure
Ollama Runtime	Model process crash or out-of-memory	FastAPI catches <code>`URLError`</code> and returns informative fallback without gateway termination.
Code Sandbox	Script compilation or runtime exception	Captures <code>`stderr`</code> and non-zero exit code, returning structured error trace to frontend.
PDF Extraction	Corrupted or password-protected PDF	PyPDF exception trapped; returns <code>`400 Bad Request`</code> with exact failure reason.
Database Sockets	Concurrent SQLite lock contention	SQLite connection per request with auto-commit and fast row-level transactions.

Chapter 12: Industrial Sizing, Deployment & Hardware Architecture

12.1 Hardware Sizing Matrix

Component	Minimum Development Spec	Recommended Industrial Production Spec
Host CPU	Intel Core i7 (8 Cores, 16 Threads)	Dual Intel Xeon Silver / Gold (32+ Cores)
System RAM	16 GB DDR4/DDR5	64 GB – 128 GB ECC Registered DDR5
GPU Acceleration	CPU Only (Quantized 4-bit)	NVIDIA RTX 4090 (24GB VRAM) or A5000 / A6000
Storage Subsystem	256 GB NVMe SSD	1 TB Enterprise PCIe 4.0 NVMe SSD (RAID-1)
Operating System	Windows 11 / Ubuntu 22.04 LTS	Ubuntu 22.04 / 24.04 LTS Server (Air-Gapped)
Network Interface	Local Loopback (127.0.0.1) Only	Dual 10GbE Isolated Refinery Intranet LAN

12.2 Verification Sign-Off & Deliverable Checklist

- [x] **Finalized Working Backend:** 11 active endpoints running on FastAPI.
- [x] **Working Database Integration:** SQLite schema (sessions, messages, documents).
- [x] **CRUD Operations Documentation:** Complete Create, Read, Update, Delete specifications.
- [x] **Database Schema & ER Diagram:** 1-to-many relationship documented.
- [x] **Frontend-Backend Integration:** Sequence diagrams and multi-step ReAct agent flows.
- [x] **API Testing Results & Proof Screenshots:** 12/12 passing unit & integration tests.
- [x] **Air-Gap Verification:** Real-time socket auditing confirming 0 external outbound calls.
- [x] **Responsive UI Validation:** Desktop, Tablet, and Mobile viewport compatibility.